photosynthesis. Animals generally have to work harder to find their food—they have to move, and this uses extra resources. Optimal foraging theory (OFT) proposes that animals try to gather resources in the most efficient way to avoid using additional energy. Searching for and capturing food takes energy and time. The animal needs to gain maximum benefit for minimal effort in order to achieve optimal fitness. OFT helps predict the best strategy that an animal can use to achieve this goal.

“Diets should be broad when prey are scarce, but narrow if food is abundant.”

Eric Pianka

Foraging theories

The first theory of foraging by wild animals did not emerge until the mid-1960s, when Americans Robert MacArthur and Eric Pianka examined the question of why, when a range of food was available to them, animals often restricted themselves to a few preferred types of prey. They argued that natural selection favored animals whose behavior maximized their net energy intake per unit of time spent foraging. An animal’s foraging time includes searching for prey and the killing and eating of the food (handling time).

These ideas were developed by American ecologists H. Ronald Pulliam and Eric Charnov and Australian ecologist Graham Pyke. It seems that OFT works best for mobile foragers seeking immobile prey, and some researchers believe it is less relevant when prey are mobile.

Key choices

Animals must choose which types of food to eat, which is rarely straightforward. For example, American ecologists Howard Richardson and Nicolaas Verbeek studied Northwestern Crows feeding on clams in the